Mingi Kang

Brunswick, ME 04011, USA
+1 (818) 795-2054 | mkang2@bowdoin.edu | mingikang31@gmail.com
GitHub | LinkedIn | Academic Portfolio | Personal Website

EDUCATION

Bachelor of Arts in Computer Science and Minor in Mathematics — Expected May 2026
Bowdoin College, ME
GPA: 3.67/4.00

Study Abroad Semester, Computer Science — August - December 2024
Aquincum Institute of Technology, Budapest, Hungary

RESEARCH INTERESTS

I am interested in developing deep architectures that integrate visual, linguistic, and structured data. Through my research experiences, from computational cognitive modeling to computer vision to medical imaging, I've explored how different modalities can inform and enhance each other. I'm particularly drawn to attention mechanisms, cross-modal alignment techniques, and self-supervised learning approaches that enable models to reason across modalities. I seek to continue exploring multimodal architectures with applications in healthcare, scientific computing, and HCI.

RESEARCH METHODS

Computer Vision ∗ Computational Imaging ∗ Deep Learning ∗ Machine Learning ∗ Natural Language Processing ∗ Statistical Analysis ∗ Data Science ∗ Experimental Design ∗ Data Collection & Management

TECHNICAL SKILLS

Deep Learning Frameworks: PyTorch, TensorFlow, Keras
Machine Learning/Data Science Libraries: Scikit-Learn, NumPy, Pandas, Matplotlib, SciPy
Programming Languages:

- *Proficient*: Python, R, Java
- *Intermediate*: TypeScript, JavaScript, Kotlin, SQL, C, Latex, Markdown

RELEVANT COURSEWORK

Computer Science:

- Artificial Intelligence, Computer Systems, Algorithms, Data Structures, Software Engineering, Mobile Software Engineering, Cryptography, Data Science, Computational Game Theory (current)

Mathematics:

- Linear Algebra, Multivariable Calculus, Probability, Statistics, Mathematical Reasoning, Advanced Topics in Probability and Statistics (current)

PUBLICATIONS AND PRESENTATIONS

Publications

1. **Kang M.**, Farias. J. (2025). Attention via Convolutional Nearest Neighbor. *In preparation for IEEE ICASSP submission*.
2. Kumar A.A., **Kang M.**, Kronenberger W.G., Jones M.N., Pisoni D. (2024). Structures and process-level lexical interactions in memory search: A case study of individuals with cochlear implants and normal hearing. In *Proceedings of the 46th Annual Meeting of the Cognitive Science Society (Vol. 46)*. DOI: https://escholarship.org/uc/item/7vn9q9hh

Oral Presentation

1. **Kang M.** (July 2025). Parallel qMRI Reconstruction from 4x Accelerated Acquisitions. *McKelvey School of Engineering Summer Symposium,* St. Louis, Missouri.

Poster Presentations

1. **Kang M.** (July 2025). Parallel qMRI Reconstruction from 4x Accelerated Acquisitions. *McKelvey School of Engineering Poster Palooza,* St. Louis, Missouri.
2. **Kang M.**, Kumar A. (July 2024). Structure and process-level lexical interactions in memory search: A case study of individuals with cochlear implants and normal hearing. *Annual Conference of the Cognitive Science Society 2024*, Rotterdam, Netherlands.

RESEARCH EXPERIENCE

NAME OF PROJECT, Bowdoin College, Brunswick, Maine August 2025 – Present
Honors Thesis Project
Advisor: Jeova Farias

- TBD
-

Computational Imaging Group, Washington University in St. Louis, Missouri May 2025 – August 2025
McKelvey Summer Engineering Fellow
Advisor: Ulugbek Kamilov, Shirin Shoushtari

- Developed supervised learning extension of SPICER framework incorporating clinical patient qMRI datasets from Washington University Medical School for enhanced reconstruction accuracy.
- Engineered efficient model compression by redesigning Deep Unfolding U-Net from SPICER into streamlined non-cascading U-Net and Attention U-Net architectures with integrated automatic coil sensitivity map estimation, achieving 4x parameter reduction while maintaining comparable reconstruction performance.
- Created normalization techniques including ACS region-specific and coil-instance normalization for under-sampled k-space measurements preprocessing.
- Achieved reconstruction performance of 37 dB PSNR and 0.923 SSIM using U-Net with Coil Sensitivity Map Estimation module.
- Collaborated with medical researchers at Washington University Medical School for experimental result validation of reconstructed qMRI outputs.

Farias Lab, Bowdoin College, Brunswick, Maine January 2025 - May 2025
Research Assistant
Advisor: Jeova Farias

- Designed novel *Convolutional Nearest Neighbor Attention (ConvNN Attention)* algorithm featuring hard selection of K-Nearest Neighbor tokens through norm-based similarity metrics integrated with convolutional layers.
- Implemented comprehensive PyTorch modules for ConvNN Attention with ConvNN expansion and linear projection, compatible with both standard Transformer and Vision Transformer architectures.
- Demonstrated significant performance gains of 5-10% improvement over standard attention mechanisms on CIFAR10 and CIFAR100 classification benchmarks.
- Preparing research manuscript on *ConvNN* and *ConvNN Attention* methodologies for submission to IEEE International Conference on Acoustics, Speech, and Signal Processing (Fall 2025).

Farias Lab, Bowdoin College, Brunswick, Maine January 2024 - August 2024
Christenfeld Summer Research Fellow
Advisor: Jeova Farias

- Developed *Convolutional Nearest Neighbor (ConvNN)* algorithm that enhances spatial learning by incorporating norm-based K-Nearest Neighbor pixel selection into standard convolution operations.
- Build modular PyTorch implementations of 1D and 2D *ConvNN* layers featuring random/spatial pixel sampling, pixel-shuffling, and coordinate encoding variations.
- Achieved substantial performance improvements of 5-8% over standard convolution for CIFAR10 classification within All Convolutional Network architecture, demonstrating enhanced spatial feature learning.

Lexicon Lab, Bowdoin College, Brunswick, Maine December 2022 - May 2024
Research Assistant
Advisor: Abhilasha Kumar

- Conducted computational cognitive science research investigating semantic, phonological, and frequency-based lexical processes in memory search, comparing neurotypical and prelingually deaf populations with cochlear implants.
- Executed behavioral experiments using verbal fluency tasks in animal semantic domains, utilizing analytical frameworks for within- and between-cluster transitions in cognitive search behavior.
- Improved computational cognitive models integrating semantic representations (word2vec, speech2vec embeddings) with memory search theories using Python package *Forager* for quantitative analysis of memory search.
- Discovered differential emphasis on speech-derived representations between cochlear implant and normal hearing participants, overturning initial research hypotheses.

Ungated Research, Bowdoin College, Brunswick, Maine April 2024 - Present
Research Assistant
Advisor: Martin Abel

- Architected and maintained centralized research platform providing open access to working papers from leading economic journals, serving academic researchers globally.
- Expanded international accessibility features by implementing automated Spanish, French, and Chinese summary translations while optimizing database schemas for seamless integration of new journal sources.
- Developed automated data pipeline integrating Google Sheets API with MariaDB and Amazon S3 infrastructure, achieving 85% reduction in journal upload processing time and significantly improving system scalability.

TEACHING & MENTORING EXPERIENCE

Learning Assistant, Bowdoin College, Brunswick, Maine Spring/Fall 2025
Course: Introduction to Computer Science

- Conduct weekly office hours providing guidance on course material, coding assignments, and project development to reinforce core computer science concepts.
- Support student career development through insights on technology pathways, research opportunities, and internship preparation.

Sophomore Bootcamp Leader, Bowdoin College, Brunswick, Maine January 2025
Career Exploration and Development Group

- Led workshops for sophomores on resume writing, interview techniques, and navigation of internship recruitment in technology and research fields.
- Mentored participants in Bowdoin's inaugural Sophomore Bootcamp Hackathon, guiding full-cycle development of end-to-end projects while supporting technical and problem-solving skill development.

AWARDS AND HONORS

Best Poster Presentation in the McKelvey Engineering Summer Research Poster Competition, WashU, $100 July 2025
McKelvey Engineering Summer Research Fellowship, Washington University in St. Louis $7,200 May 2025
Allen B. Tucker Computer Science Research Prize, Bowdoin College $50 May 2025
CRA Undergraduate Award, Last Mile Education $4,000 April 2025
NYC Stem Award, Last Mile Education $1,500 March 2025
Google AI 2024 Award, Last Mile Education $595 January 2025
Christenfeld Summer Research Fellowship, Bowdoin College $4,800 April 2024
NSF Student Faculty Research Fellowship, Bowdoin College $1,800 January 2024
National College Match Finalist, QuestBridge September 2021

LANGUAGES

English (Native) ∗ Korean (Native) ∗ German (Conversational)

PERSONAL INTERESTS

Snowboarding ∗ Cooking ∗ Weightlifting ∗ Matcha/Coffee ∗ Films/Shows ∗ Traveling ∗ Pottery ∗ Baseball

PERSONAL ACCOMPLISHMENTS

- 215 lbs. Max Bench Press at (160 lbs. bw)
- 6:13 Mile Run
- Dinner party of 12 people, serving as primary chef